

Free End-to-End Data Engineering Project Roadmap

This project is designed to help you practice the exact tools commonly asked in data engineering interviews:

- PySpark
- Airflow
- Dagster (optional but valuable)
- PostgreSQL
- SQL
- Linux commands
- SSH
- AWS S3 concepts
- ETL/ELT pipelines
- Data warehousing concepts
- Incremental loading
- Scheduling and orchestration

The goal is to build everything for free on your laptop.

1. Project Idea

Project: E-Commerce Analytics Pipeline

You will simulate a company collecting e-commerce order data.

Pipeline flow:

CSV/API Data -> S3-like Storage -> PySpark Transformations -> PostgreSQL Warehouse -> Airflow/Dagster Orchestration -> Analytics SQL Queries

This looks extremely good on a resume because it covers:

- Batch pipelines
 - Spark transformations
 - Scheduling
 - Cloud concepts
 - Warehousing
 - SQL analytics
 - Linux operations
 - Data modeling
-

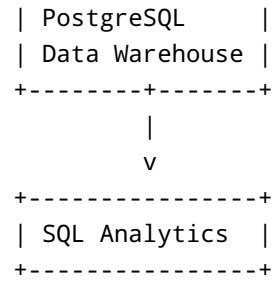
2. Recommended Tech Stack (100% Free)

Component	Tool
Programming	Python
Big Data Processing	PySpark
Workflow Orchestration	Airflow
Optional Modern Orchestrator	Dagster
Database	PostgreSQL
S3 Simulation	MinIO
Containerization	Optional Docker
Linux Environment	WSL Ubuntu
SQL Tool	DBeaver
SSH Practice	OpenSSH
Version Control	Git + GitHub

Since you prefer avoiding Docker for performance, this setup works locally.

3. Architecture

```
+-----+
| Raw CSV Files |
+-----+
      |
      v
+-----+
| MinIO (S3)    |
+-----+
      |
      v
+-----+
| PySpark ETL   |
| Cleaning      |
| Aggregations  |
+-----+
      |
      v
+-----+
```



Airflow/Dagster schedules and monitors the pipeline.

4. Core Features You Should Build

Stage 1 — Raw Data Ingestion

Use:

- Kaggle e-commerce datasets
- CSV files
- Fake API data

Tasks:

- Store raw files in MinIO bucket
- Organize folders by date
- Practice Linux file movement commands

Important Linux Commands:

```
ls
cd
mkdir
mv
cp
rm
find
grep
cat
tail
head
chmod
ps
kill
```

5. Setup Linux Environment

Install:

- WSL Ubuntu
- Python
- Java (required for Spark)
- PostgreSQL
- OpenSSH

Useful interview-level commands:

```
sudo apt update
sudo apt install
systemctl status
ps aux
netstat -tulnp
ssh
scp
nano
vim
```

6. SSH Practice

SSH is often asked indirectly in interviews.

You can simulate SSH locally.

Install SSH Server

```
sudo apt install openssh-server
```

Start Service

```
sudo service ssh start
```

SSH Into Local Machine

```
ssh username@localhost
```

Practice:

- File transfer using scp
- SSH keys
- Passwordless login
- Linux navigation remotely

Example:

```
scp orders.csv username@localhost:/home/username/data/
```

7. Setup PostgreSQL

Create tables:

```
CREATE TABLE orders (  
    order_id INT,  
    customer_id INT,  
    order_date DATE,  
    product_id INT,  
    amount NUMERIC  
);
```

Practice:

- Indexes
- Joins
- Window functions
- Incremental loads
- Partitioning concepts

Important interview SQL:

```
ROW_NUMBER()  
RANK()  
DENSE_RANK()  
LAG()
```

```
LEAD()  
CTEs  
CASE WHEN  
GROUP BY  
HAVING
```

8. PySpark Pipeline

Install

```
pip install pyspark
```

What Your Pipeline Should Do

Read Raw CSV

```
spark.read.csv()
```

Clean Data

- Remove nulls
- Cast types
- Deduplicate
- Standardize columns

Transform

- Revenue aggregations
- Customer segmentation
- Daily sales
- Product trends

Write to PostgreSQL

Use JDBC.

Important concepts:

- Lazy evaluation
- Transformations vs actions
- Partitioning
- Repartition vs coalesce

- Spark DAG

These are very common interview topics.

9. Add Airflow

Why?

Companies LOVE orchestration experience.

Airflow lets you:

- Schedule jobs
- Monitor failures
- Retry tasks
- Build DAGs

Install

```
pip install apache-airflow
```

DAG Example Flow

```
Fetch CSV
↓
Upload to MinIO
↓
Run Spark ETL
↓
Load PostgreSQL
↓
Run Validation SQL
```

Key interview concepts:

- DAG
- Operators
- Scheduling
- Retries
- Dependencies
- Backfills
- Sensors

- XCom
-

10. Add Dagster (Optional)

Dagster is modern and cleaner than Airflow.

You can mention:

- You know traditional orchestration (Airflow)
- You explored modern orchestration (Dagster)

That stands out.

Good enough:

- One simple pipeline
 - Asset-based workflow
 - Logging
-

11. Simulate AWS S3 for Free

Use MinIO.

It behaves like S3 locally.

You can say in interviews:

“I used MinIO locally to simulate AWS S3 APIs.”

This is completely acceptable for portfolio projects.

Important S3 concepts:

- Buckets
 - Object storage
 - Partition folders
 - Lifecycle concepts
 - Parquet storage
-

12. Add Incremental Loading

Very important for interviews.

Example:

- Only process new files
- Track latest processed timestamp
- Upsert into PostgreSQL

Concepts:

- Watermarking
 - CDC basics
 - Incremental ETL
 - Idempotency
-

13. Data Modeling

Create:

Fact Table

```
fact_orders
```

Dimension Tables

```
dim_customers  
dim_products  
dim_date
```

Learn:

- Star schema
- Surrogate keys
- OLTP vs OLAP

Very commonly asked.

14. Recommended Folder Structure

```
project/
|
├─ data/
├─ dags/
├─ spark_jobs/
├─ sql/
├─ logs/
├─ configs/
├─ scripts/
├─ tests/
└─ README.md
```

15. BEST Resume Points

After building this project, you can honestly write:

- Built end-to-end ETL pipeline using PySpark and PostgreSQL
- Implemented workflow orchestration using Airflow DAGs
- Simulated AWS S3 storage using MinIO
- Developed incremental batch processing pipelines
- Performed SQL-based analytics and warehousing
- Automated Linux-based data operations using shell scripting
- Configured SSH-based remote data transfer workflows

16. Important Interview Topics This Project Covers

PySpark

- Lazy evaluation
- Partitioning
- Shuffles
- Transformations/actions
- Joins
- Optimizations

SQL

- Window functions

- Aggregations
- Joins
- Indexes
- Query optimization

Airflow

- DAGs
- Scheduling
- Retries
- Sensors

Linux

- Process management
- Permissions
- File operations
- Cron jobs

Data Engineering

- ETL vs ELT
 - Batch processing
 - Incremental loading
 - Data warehousing
 - Orchestration
-

17. Suggested Timeline

Week 1

- Linux setup
- PostgreSQL
- Basic SQL
- SSH

Week 2

- PySpark basics
- Read/write CSV
- Data cleaning

Week 3

- PostgreSQL integration
- Aggregations
- Incremental loading

Week 4

- Airflow DAGs
- Scheduling
- Monitoring

Week 5

- MinIO S3 simulation
- Logging
- Better architecture

Week 6

- README
- GitHub polish
- Add diagrams
- Add interview notes

18. GitHub README Sections

Your README should contain:

- Architecture diagram
- Tech stack
- Setup steps
- DAG screenshots
- Sample SQL queries
- Folder structure
- Pipeline explanation
- Challenges faced
- Future improvements

A good README massively improves resume impact.

19. Final Recommendation

Do NOT try to build a massive production-level system immediately.

First build:

1. Working PySpark ETL
2. PostgreSQL loading
3. Airflow orchestration
4. MinIO integration
5. Incremental loading

Then improve step-by-step.

This project alone is enough to discuss for:

- Junior Data Engineer
- Data Analyst Engineer
- Analytics Engineer
- Big Data Intern
- ETL Developer

if explained properly.

20. Best Free Dataset Sources

- Kaggle
- NYC Taxi Dataset
- Instacart Dataset
- Amazon Reviews Dataset
- Fake Store API
- GitHub public datasets

Recommended starting dataset:

Small e-commerce orders dataset.

Much easier to finish and explain in interviews.